EXPERIMENT 4

Single Page Application using React Router

AIM

To develop a Single Page Application using React Router for navigation between components without reloading the page.

ALGORITHM

1. Create a React application and install react-router-dom.
2. Create Home, About and Contact components.
3. Wrap the application with BrowserRouter.
4. Define routes using Routes and Route.
5. Use Link components to navigate and test each page.

PROGRAM

```
// Install: npm install react-router-dom
// App.jsx
import {BrowserRouter,Routes,Route,Link} from 'react-router-dom';

function Home(){return <h2>Home Page</h2>}
function About(){return <h2>About Page</h2>}
function Contact(){return <h2>Contact Page</h2>}

export default function App(){
 return <BrowserRouter>
  <nav>
   <Link to="/">Home</Link> |
   <Link to="/about">About</Link> |
   <Link to="/contact">Contact</Link>
  </nav>
  <Routes>
   <Route path="/" element={<Home/>}/>
   <Route path="/about" element={<About/>}/>
   <Route path="/contact" element={<Contact/>}/>
  </Routes>
 </BrowserRouter>;
}
```

RESULT

Thus, a React Single Page Application with component-based navigation was created successfully.

VIVA QUESTIONS

1. What is a Single Page Application?
2. What is React Router?
3. What is a component in React?
4. What is the use of BrowserRouter?
5. How does client-side routing avoid page reloads?